

DISTRIBUTED REAL-TIME CONTROL SYSTEMS

FIRST STAGE



TÉCNICO LISBOA

Real-Time Cooperative Control of a Distributed Illumination System

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1 Introduction

This report presents the first stage of a distributed illumination control system. A local controller was implemented on the Raspberry Pi Pico for one luminaire, including measurement, actuation, calibration, control, disturbance handling, and a serial interface for testing.

2 Measurement and Calibration

The illuminance was measured with the LDR connected to A0. To reduce ADC noise, several samples were averaged before converting the reading into voltage, resistance, and illuminance, according to Equations (1)–(4).

Since the two luminaires have different sensor responses, the corresponding calibration parameters m and b , obtained in the second session, were selected at startup through ID.PIN. Automatic calibration was also used to estimate background lux at $PWM = 0$ and the maximum illuminance at $PWM = 4095$, allowing the useful static gain to be computed using Equation (5).

3 Local Controller Implementation

3.1 Control Signal and Actuation

The LED is driven by a 12-bit PWM signal in the range 0–4095. The control output is saturated before conversion into PWM units, ensuring valid actuation. A PWM frequency of 60 kHz was used to reduce the effect of switching noise on the LDR measurement.

3.2 Feedforward

A feedforward term was used to estimate the duty cycle required for a given illuminance reference from the calibrated static gain, as described in Equation (6). This provides an approximate nominal command, so that feedback only needs to correct residual errors and disturbances.

3.3 Feedback PID

The controller was implemented as a 2DOF PID updated every $T_s = 0.01$ s. At each sampling

instant, it uses the illuminance reference, the filtered measured illuminance, and the feedforward term to compute a normalized control signal.

Set-point weighting was introduced in the proportional action with $\beta = 0.20$, while the integral action uses the full tracking error. This reduces the control effort during reference changes without significantly affecting steady-state accuracy. The gains were defined from the calibrated static gain as $K_p = 0.35/\text{gain}$, $K_i = 0.4/\text{gain}$, and $K_d = 0$, so that the tuning scales with each luminaire. Derivative action was not used because it increased sensitivity to measurement noise without relevant improvement in the transient response.

3.4 Anti-Windup

To prevent excessive integral accumulation when the control signal saturates, anti-windup was implemented through back-calculation, as described in Equations (12) and (13). This improves recovery from saturation and shortens the transient after large reference changes.

3.5 Filtering and Disturbance Handling

The measured illuminance was filtered before being used by the controller, according to Equation (15). Different values of α were used to obtain a faster response to commanded reference changes and a smoother response to external disturbances.

A disturbance was detected when the reference remained approximately constant but the measured illuminance changed abruptly, as described in Equation (16). In that case, the controller entered a slower mode, with reduced filter coefficient and limited PWM variation, according to Equation (17). This avoided overly aggressive corrections to external light changes and helped reduce visible flicker. The values of α used in each condition are summarized in Table 2.

4 Serial Interface

A serial interface was essential during development, since it enabled direct testing, tuning, and experimental validation of the controller. The implemented commands are summarized in Table 3.

5 Experimental Results

5.1 Steady-State Characteristic of the System

Figure 4 shows the steady-state PWM–illuminance characteristic of the luminaire. The relation is approximately monotonic and almost linear over most of the useful operating range, with some deviation at low PWM values. This result supports the use of a static gain estimate for initialization of the feedforward term.

5.2 Step Responses Under Different Illuminance Conditions

Figures 5, 6, and 7 show the step responses under different illuminance conditions. In all cases, the measured illuminance follows the reference with low error and limited overshoot.

At low illuminance levels, the response is slower due to system nonlinearity, as shown in Figures 20 and 21 and confirmed by Table 6, where the $3 \rightarrow 9$ lux transition is faster than the $0 \rightarrow 3$ lux case. At higher levels, Figure 7 shows stable tracking without significant oscillations.

5.3 Response to External Disturbances

Figures 8 and 9 show the response to external disturbances. In both cases, the controller compensates the disturbance progressively through the PWM command, with a smoother recovery than in commanded reference changes. This behaviour is consistent with the implemented disturbance mode and helps reduce flicker.

5.4 Steady-State Operation

Figures 10 and 11 show accurate steady-state regulation at two operating points, with measured illuminance close to the reference and a nearly constant control signal. This is confirmed by Tables 4 and 5, which show very small steady-state errors: 0.0048 lux at 5 lux and 0.0276 lux at 15 lux.

5.5 Illustration of PID Features

Figures 12 and 13 show that both responses converge to the same reference, but with anti-

windup enabled the controller reaches that value earlier and settles faster. Although the difference is not large, the transient is clearly shorter with anti-windup.

Figures 14 and 15 show that feedback is essential for accurate tracking. Without feedback, the measured illuminance remains visibly offset from the reference, whereas with feedback enabled the tracking error becomes much smaller and the output follows the reference more closely.

Figure 17 shows that $\beta = 1$ and $\beta = 0.2$ lead to similar illuminance responses, while the main difference appears in the transient control signal. This confirms that set-point weighting mainly reduces control effort during reference changes, with little effect on the output response.

5.6 Sampling Jitter

Figures 18 and 19, together with Table 7, summarize the sampling-period jitter of the control loop. Over 13000 samples, the sampling period remained centered around the nominal $10000 \mu\text{s}$ (100 Hz), with mean $10000.000 \mu\text{s}$, standard deviation $1.516 \mu\text{s}$, and maximum absolute deviation $80 \mu\text{s}$, indicating stable average timing with occasional small deviations.

5.7 Performance Metrics

The implemented solution was evaluated using the performance metrics defined in the project statement: accumulated energy, visibility error, and flicker error. These quantities were computed from Equations (18), (19), (20), and (21). Following the project guidelines, the visibility metric was computed using the illuminance estimated from the LED command rather than from the instantaneous LDR measurement.

Figures 22, 23, and 24 show that the accumulated energy evolves smoothly over time, while both visibility error and flicker remain low during most of the experiment. This indicates that the controller keeps the desk illuminance close to the required level with smooth control action and without excessive oscillatory actuation.

These results are consistent with Tables 4, 5, and 6, which confirm small steady-state errors, limited overshoot, and accurate local control.

A Experimental Figures



Figure 1: Exterior view of the box used as the scaled office model.



Figure 2: Interior view of the box showing the illuminated area and the placement of the luminaire.

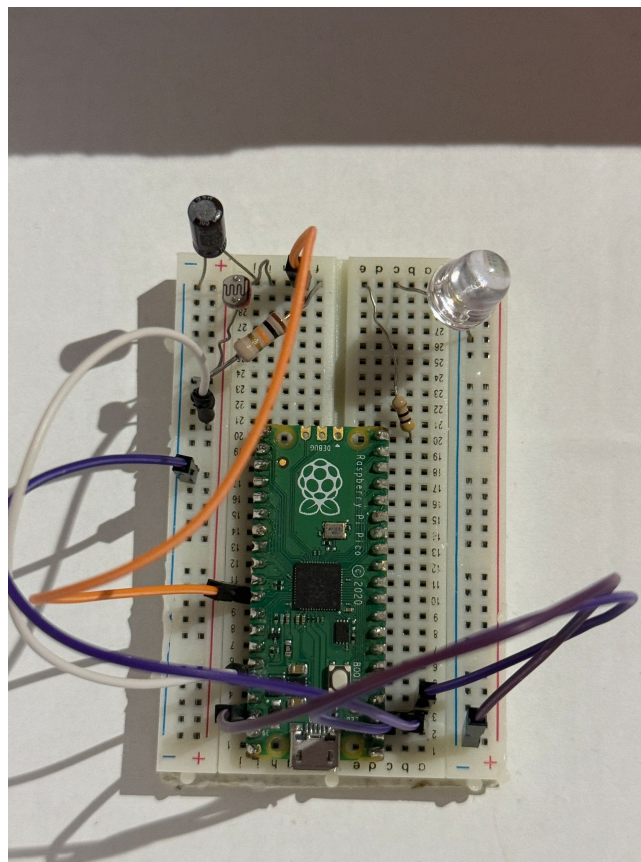


Figure 3: Prototype of the local luminaire node, including the Raspberry Pi Pico, LED driver, and LDR measurement circuit.

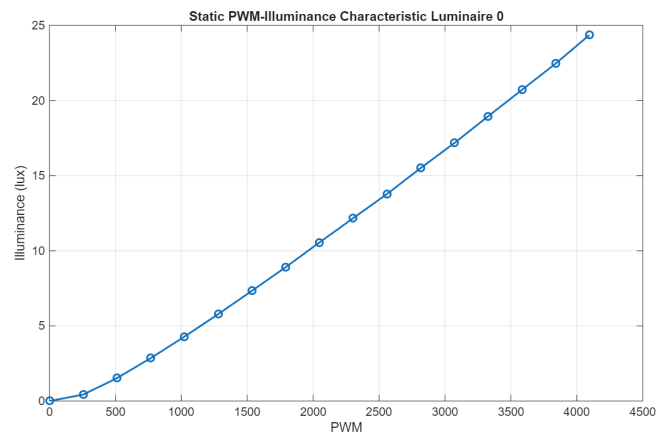


Figure 4: Static PWM–illuminance characteristic of the luminaire

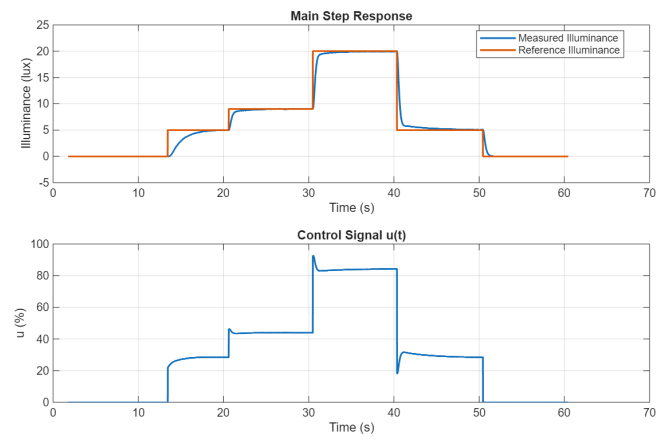


Figure 5: Main step response of the local controller.

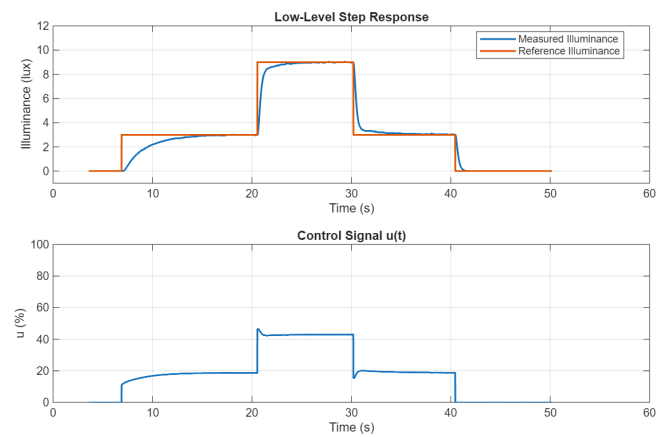


Figure 6: Low-illumination step response of the local controller.

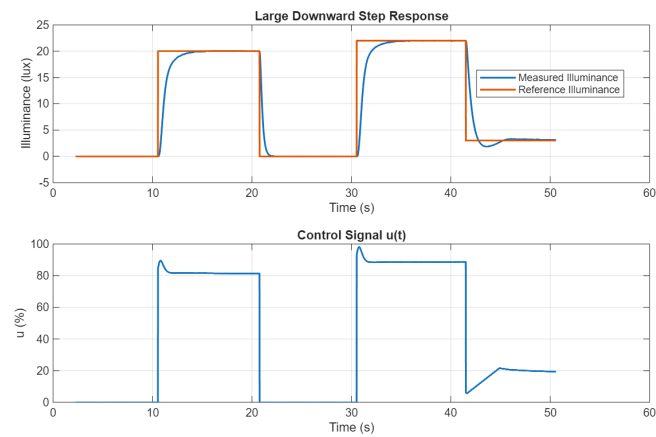


Figure 7: High-illuminance step response of the local controller.

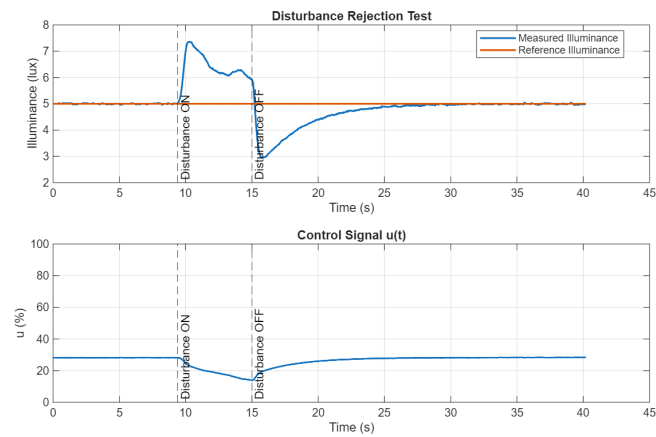


Figure 8: Response to an external disturbance under a 5 lux reference.

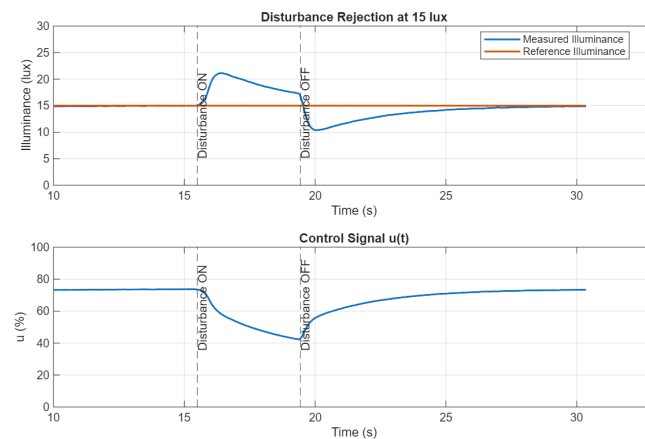


Figure 9: Response to an external disturbance under a 15 lux reference.

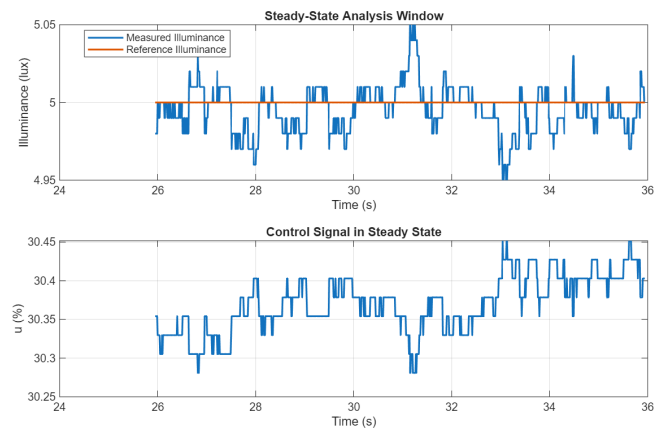


Figure 10: Steady state analysis at 5 lux

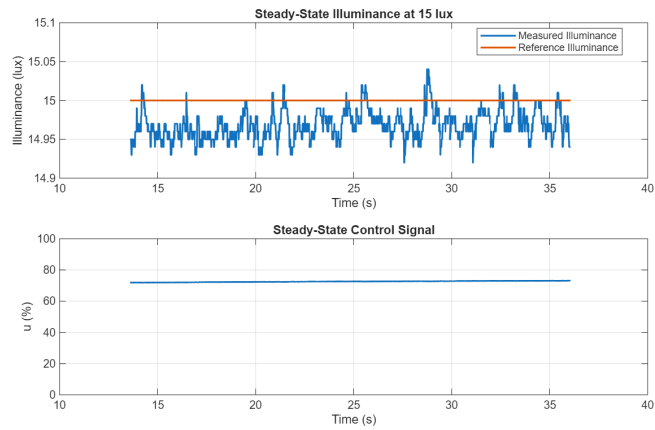


Figure 11: steady state analysis at 15 lux

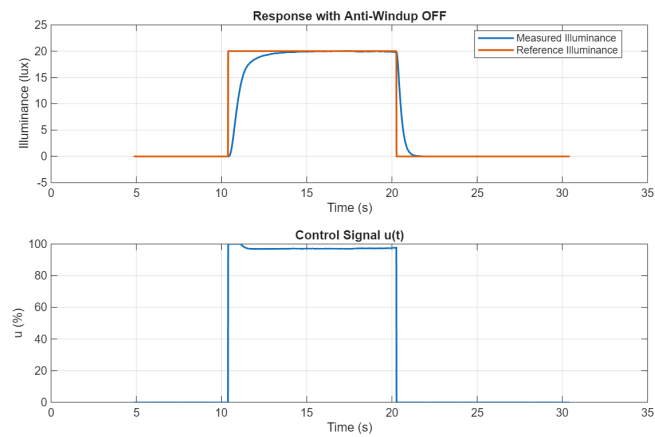


Figure 12: Anti-windup off

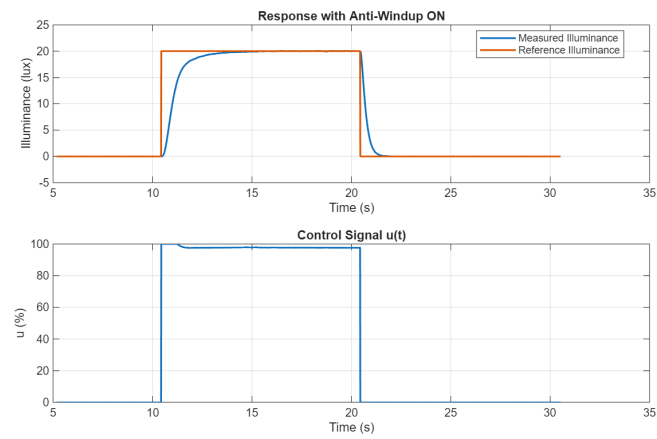


Figure 13: Anti-windup on

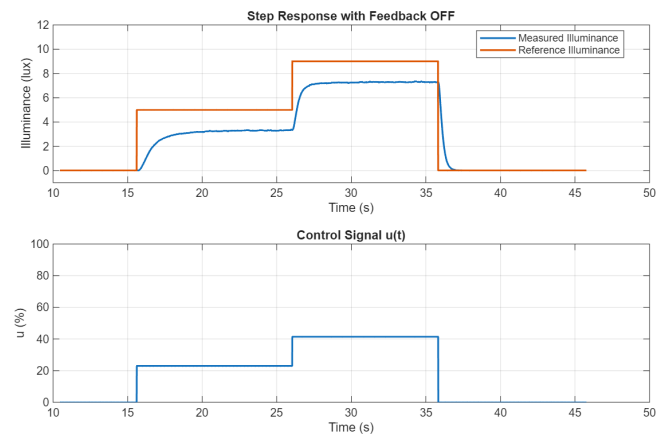


Figure 14: Feedback off

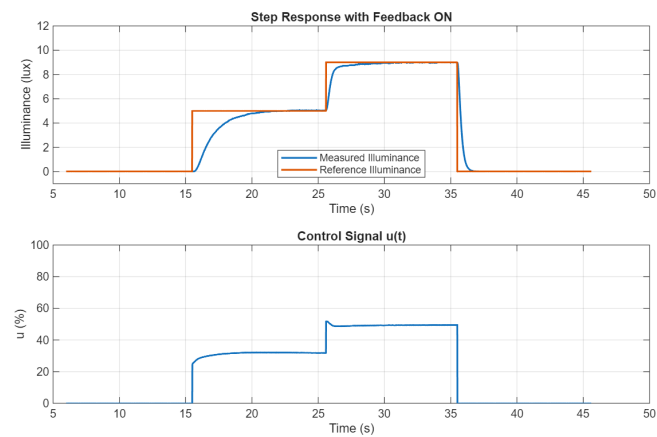


Figure 15: Feedback on

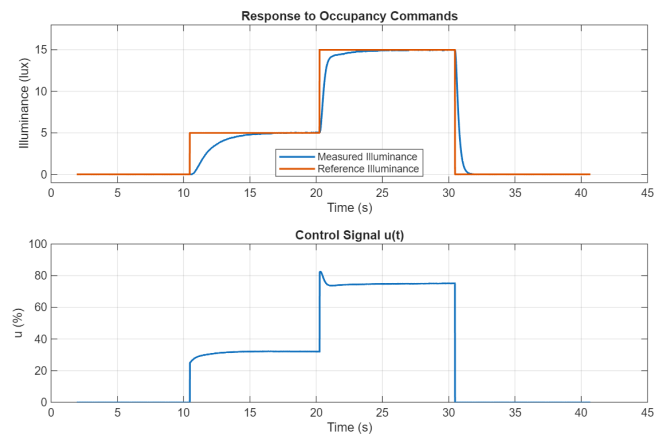


Figure 16: Response to occupancy-state commands.

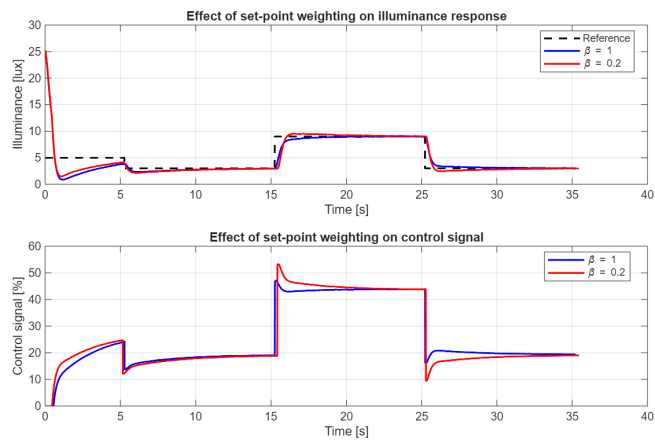


Figure 17: Set-point weighting

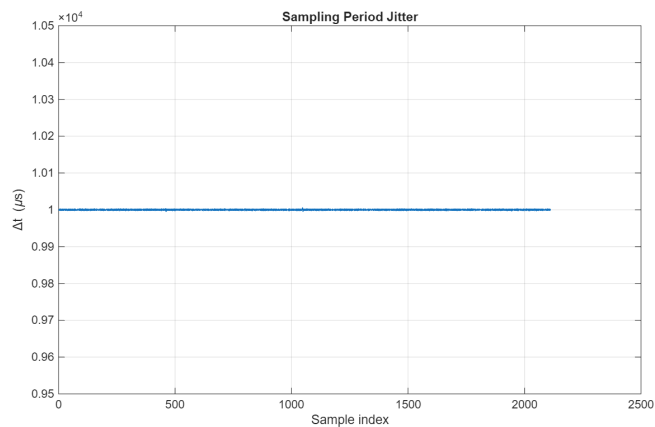


Figure 18: Sampling jitter

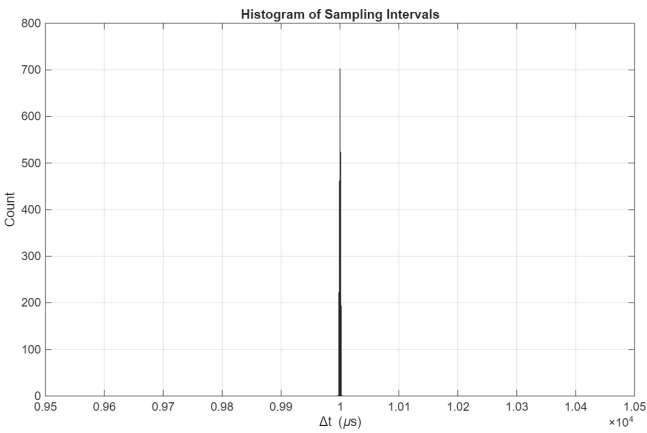


Figure 19: Histogram jitter

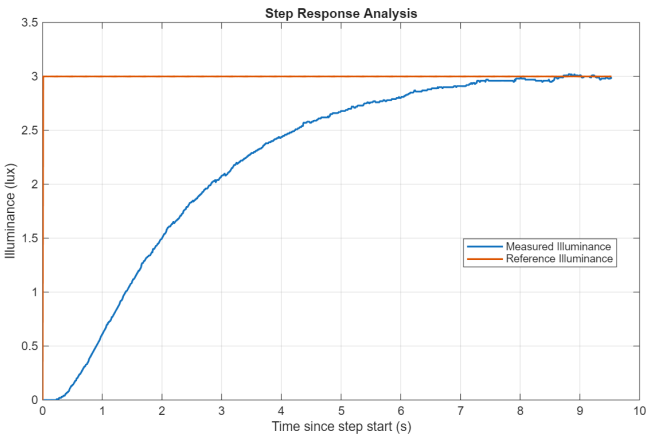


Figure 20: Step response 0-3

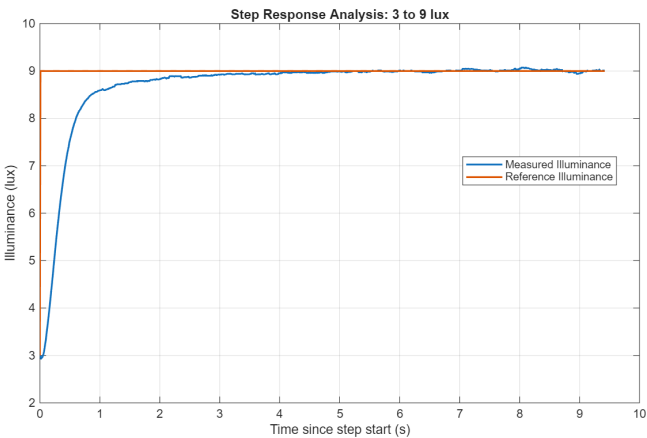


Figure 21: Step response 3-9

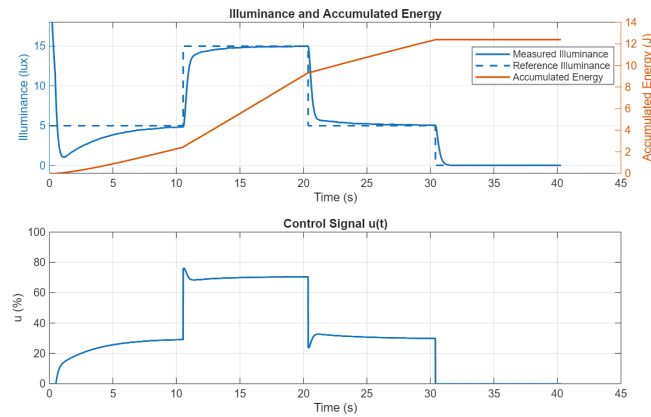


Figure 22: Evolution of the accumulated energy during the experiment. The upper plot shows the measured illuminance, the reference illuminance, and the accumulated energy, while the lower plot shows the control signal in percentage.

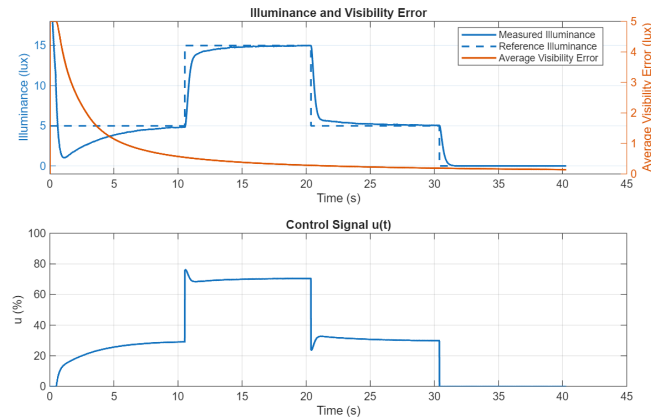


Figure 23: Evolution of the average visibility error during the experiment. The upper plot shows the measured illuminance, the reference illuminance, and the average visibility error, while the lower plot shows the control signal in percentage.

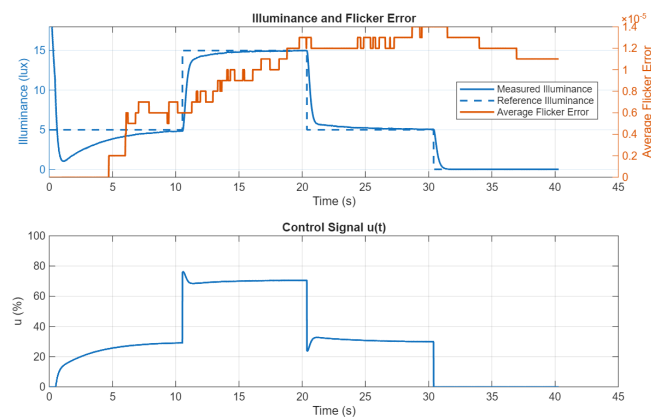


Figure 24: Evolution of the average flicker error during the experiment. The upper plot shows the measured illuminance, the reference illuminance, and the average flicker error, while the lower plot shows the control signal in percentage.

B Main Equations

B.1 Measurement and Calibration

$$V_{LDR} = \frac{ADC}{4095} \cdot 3.3 \quad (1)$$

$$R_{LDR} = R_{fixed} \frac{3.3 - V_{LDR}}{V_{LDR}} \quad (2)$$

$$lux_{raw} = 10^{\frac{\log_{10}(R_{LDR}) - b}{m}} \quad (3)$$

$$lux = \max(lux_{raw} - backgroundLux, 0) \quad (4)$$

$$gain = maxLux - backgroundLux \quad (5)$$

B.2 Control Law

$$u_{ff} = \text{sat} \left(\frac{refLux}{gain} \right) \quad (6)$$

$$e_p(k) = \beta r(k) - y(k) \quad (7)$$

$$u(k) = u_{ff}(k) + u_{fb}(k) \quad (8)$$

$$e(k) = r(k) - y(k) \quad (9)$$

$$u_{fb}(k) = K_p e_p(k) + I(k) \quad (10)$$

$$u_{sat}(k) = \text{sat}(u(k), 0, 1) \quad (11)$$

$$aw(k) = K_{aw}(u_{sat}(k) - u(k)) \quad (12)$$

$$I(k+1) = I(k) + T_s (K_i e(k) + aw(k)) \quad (13)$$

$$K_p = \frac{0.35}{gain}, \quad K_i = \frac{0.4}{gain}, \quad K_d = 0 \quad (14)$$

B.3 Filtering and Disturbance Handling

$$y_f(k) = (1 - \alpha) y_f(k-1) + \alpha y(k) \quad (15)$$

$$|refLux(k) - refLux(k-1)| < 0.1 \quad \text{and} \quad |lux(k) - lux(k-1)| > 2.0 \quad (16)$$

$$|PWM(k) - PWM(k-1)| \leq 2 \quad (17)$$

B.4 Performance Metrics

$$E = P_{max} \sum_{k=1}^N d_{k-1}(t_k - t_{k-1}) \quad (18)$$

$$V = \frac{1}{N} \sum_{k=1}^N \max(0, L(t_k) - l(t_k)) \quad (19)$$

$$f_k = \begin{cases} |d_k - d_{k-1}| + |d_{k-1} - d_{k-2}|, & \text{if } (d_k - d_{k-1})(d_{k-1} - d_{k-2}) < 0 \\ 0, & \text{otherwise} \end{cases} \quad (20)$$

$$F = \frac{1}{N} \sum_{k=1}^N f_k \quad (21)$$

C Supplementary Tables

Table 1: Controller parameters used in the implemented local PID.

Parameter	Value
K_p	$0.35/\text{gain}$
K_i	$0.4/\text{gain}$
K_d	0
β	0.20
K_{aw}	$3K_i$
T_s	0.01 s
$u_{\min}$	0
$u_{\max}$	1

Table 2: Adaptive filtering coefficients used in the controller.

Operating condition	Filter coefficient α
Reference change	0.25
Normal operation	0.05
Disturbance mode	0.02

Table 3: Implemented serial commands for local node interaction.

Command	Description
u i val	Set PWM command of luminaire i
g u i	Read PWM command of luminaire i
r i val	Set illuminance reference of luminaire i
g r i	Read illuminance reference of luminaire i
g y i	Read measured illuminance of luminaire i
g v i	Read LDR voltage of luminaire i
o i val	Set occupancy state of luminaire i (o, l, h)
g o i	Read occupancy state of luminaire i
f i val	Enable or disable feedback control
g f i	Read feedback state
a i val	Enable or disable anti-windup
g a i	Read anti-windup state
g E i	Read accumulated energy
g V i	Read average visibility error
g F i	Read average flicker error
g d i	Read estimated external illuminance
g p i	Read instantaneous electrical power

Table 4: Steady-state performance measures at 5 lux.

Metric	Value
Reference illuminance	5.0000 lux
Mean measured illuminance	4.9952 lux
Steady-state error	0.0048 lux
Absolute steady-state error	0.0048 lux
Standard deviation (lux)	0.0150 lux
Minimum illuminance	4.9500 lux
Maximum illuminance	5.0500 lux
Mean control signal	30.3698 %
Standard deviation of $u(t)$	0.0358 %
Minimum control signal	30.2808 %
Maximum control signal	30.4518 %

Table 5: Steady-state performance measures at 15 lux.

Metric	Value
Reference illuminance	15.0000 lux
Mean measured illuminance	14.9724 lux
Steady-state error	0.0276 lux
Absolute steady-state error	0.0276 lux
Standard deviation (lux)	0.0188 lux
Minimum illuminance	14.9200 lux
Maximum illuminance	15.0400 lux
Mean control signal	72.8786 %
Standard deviation of $u(t)$	0.1410 %
Minimum control signal	72.6007 %
Maximum control signal	73.1624 %

Table 6: Performance measures of the low-level step responses.

Metric	0 → 3 lux	3 → 9 lux
Rise time (s)	4.39	0.66
Settling time (s)	7.15	2.65
Overshoot (%)	1.31	1.08
Steady-state error (lux)	0.0190	0.0095

Table 7: Sampling-period jitter statistics.

Metric	Value
Samples	13000
Mean period	10000.000 μs
Standard deviation	1.516 μs
Minimum period	9920 μs
Maximum period	10079 μs
Maximum absolute deviation	80 μs